

SWASTHIK ROSHAN M S

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Information Technology Student | Software Developer | Aspiring Data Analyst

Education

SRM Institute of Science and Technology, Tiruchirappalli

India

B.Tech. in Information Technology

Maharishi Vidya Mandir

Erode, India

Grade XII | Score: 77%

Maharishi Vidya Mandir

Erode, India

Grade X | Score: 78%

Internship Experience

Top Tech Developers

Chennai, India

Web Development Intern

- Completed a hands-on internship in **Web Development**, gaining practical exposure to modern web technologies and software development practices.
- Developed technical skills through **project-based training**, applying programming concepts to practical development tasks and demonstrating strong problem-solving abilities.
- Collaborated on development activities while strengthening skills in **web technologies, debugging, and application development**.

Projects

Smart Learning for Young Minds

- Developed a **web-based learning platform** connecting students, teachers, parents, and administrators through a unified learning environment.
- Implemented **interactive lessons, quizzes, assignments, assessments, and progress tracking** to support personalized and engaging learning.
- Designed role-based dashboards with **secure authentication, parent-teacher communication, notifications, and performance analytics** for effective learning management.

Reinforcement Learning-Based Deadlock Detection & Prevention

- Developed a **Reinforcement Learning (RL)**-based model to detect and predict deadlocks in operating system process scheduling.
- Trained the model using **simulated resource-allocation scenarios** to learn effective deadlock prevention strategies.
- Compared the proposed approach with **Banker's Algorithm**, demonstrating improved responsiveness and efficiency in simulated scheduling scenarios.

Technical Skills

Languages: Python, C, HTML, CSS

Databases: MySQL, MongoDB, Database Management Systems (DBMS)

Core CS: Object-Oriented Programming (OOP), Three-Tier Architecture, Data Structures

Web & APIs: Flask, REST APIs

Soft Skills: Problem Solving, Communication, Teamwork, Adaptability

Certifications

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** – Oracle University
- MongoDB Basics for Students** – MongoDB University
- Introduction to MongoDB (For Students)** – MongoDB University